

- The task we are solving requires random luck because we are directly deriving the answer. In other works on APO, the optimized prompts contain a lot of instructions, and without reasoning directly conclude to yes or no is difficult. For direct yes or no answers: it requires luck i.e. random search of the prompt. - my hypothesis
- "where to focus" / "where is the important thing" - based important selection of prompt parts from parents
- take the difference between child and parent as important items that improved the scores
- use LLM itself to identify the differences in the prompt between parent and child that caused the improvement
- Evaluating on the full evaluation set can be expensive (i.e. if I use reasoning). "Automatic Prompt Optimization with "Gradient Descent" and Beam Search" - solves this as multi-arm / multi-bandit problem. check that.
- 1-shot to K-shot VS showing how to solve as examples

Top approaches

- GEPA merge: Combine different best sub-modules from different systems to be single best systems
- Error taxonomy - updates the prompt only once - i think it is inspired by GRPO i.e. doing batch update rather than single update
- Evoprompt - $a + x(b - c)$
- SPO - no full validation rather they do pairwise self-judge by LLM based on outputs of different prompts
- PromptAgent - uses MCTS to explore new prompt (passes rewards to parent nodes)
- a mixed of IO and EO approach - from the Teach better or show smarter
- My idea: show trace of changes of the prompt during mutation: that will show the differences and the impact

What is necessary

- Grouping of errors - best learning
- Merging best subsystems (multi-agent)
- Show trace of changes (idea similar to: CMA-ES (Covariance Matrix Adaptation ES) , DE (used in Evoprompt), or Lamarckian & Baldwinian Evolution (classic framing) Often implemented via local search + GA, also called memetic algorithms)
- MCTS - for search
- Examples Optimization (Random and Mutation from pool)
- * Use MCTS + UCT for search (exploration + exploitation)

Step per iteration:

1. Select a system (one prompt per system)
- 2.